MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 7840

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Thin end mica window, self quenched counter tube for detection of low intensity beta radiation
See figure 1
Mounting position: Any
Weight: 7 ounces (198.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ebb	TA	Rp	Mica window thickness
Unit:	Vdc	°C	Meg	mg/cm^2
Maximum:	1,600	40	---	2.2
Minimum:	1,300	0	1	1.8
TEST CONDITIONS:	1,400	---	1	---

GENERAL:

Qualification - Required
Holding period - 60 days
Window test - See note 13
© Shelf life - Required (see quality conformance inspection, part 1)
© Acceptance test - Required (see note 16)

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
6201	Background, contamination and photosensitivity	2,6,12	t = 4 minutes (min)	N/t	---	30	Npm
Ⓒ 6211	Starting voltage	5, 12, 17	Pulse amplitude = 0.50 V; Nps = 100 (min); R2 = 1 Meg; R1 = C1 = C2 = 0; C3 = 0.01 µF	Es	---	1,250	Vdc
6216	End-of-plateau voltage	-	Ps = 3%/100 Vdc	Ee	1,650	---	Vdc
Ⓒ 6216	Relative plateau slope	16, 17	Voltage range = 1,300 to 1,600 Vdc	Ps	---	3	%/100 Vdc
6221	Response count rate and current (beta)	3,6,12	Ebb = 1,400 Vdc; counting time = 2 minutes (min)	N/t	Note 3	---	Npm
Ⓒ ---	Shelf life	15	---	---	---	---	---
	Quality conformance inspection, part 2						
6205	Leakage current	-	Ebb = 700	LIb	-	0.5	µAdc
1041	Shock	1,11	450 G	---	---	---	---
6221	Response count rate and current (hysteresis (1a))	6,7,8	Ebb = 1,400 Vdc; t = 2 minutes (min); record count rate values	---	---	---	---
6221	Response count rate and current (hysteresis (1b))	6,8	Ebb = 1,400 Vdc; t = 15 minutes (min)	---	---	---	---
6221	Response count rate and current (hysteresis (1c))	6,8	Ebb = 1,400 Vdc; t = 2 minutes (min); from reading taken from count rate (1a)	N/t	---	±20	%
6221	Response count rate and current (gamma)	6,9,12	Ebb = 1,400 Vdc; counting time = 2 minutes (min)	N/t	---	110	Npm
	Quality conformance inspection, part 3						
---	Life test	-	Group D; Ebb = 1,400 Vdc; N/t = 1,000 Npm; t = 250 hours	---	---	---	---
---	Life-test end points:						
6221	Response count rate and current (beta)	-		N/t	---	±10	%
6221	Response count rate and current (gamma)	-		N/t	---	±10	%
1031	Variable-frequency vibration	1,10,14	Ebb = 0	---	---	---	---
1026	Temperature range, operating	4,14		N/t	---	±20	%
Ⓒ ---	Packaging	18	---	---	---	---	---

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NOTES:

1. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the requirements for the following:

   a. Starting voltage.
   b. End-of-plateau voltage.
   c. Relative plateau slope.
   d. Response count rate and current (beta).
   e. Response count rate and current (gamma).

2. The tube shall be exposed to radiation from a General Electric 4-watt germicidal lamp, Type G-4T4/1, or equivalent, with relative positions as shown on figure 2.

© 3. The counting rate shall not vary by more than +0.10 N/t from the value (approximately 10,000 counts per minute (cpm)) marked on the Standard Beta Reference Source (SBRS). The SBRS shall be mounted in recess of the sample positioning shelf specified with the SBRS. The proper position of this shelf for this test is specified with the SBRS.

4. With the tube in a field giving 10,000 ±100 Npm at 1,400 Vdc, tube response (count rate) shall be determined at 1,300, 1,400, and 1,600 Vdc at each of the following temperatures and in the order shown:

   a. Room temperature.
   b. 5°C.
   c. Room temperature.
   d. 40°C.
   e. Room temperature.

5. After the holding period, starting voltage shall not vary by more than ±7 volts.

6. The tube shall be mounted vertically in the lead shield (see figure 3) of the excitation unit.

7. The tube shall have been inoperative for a 24-hour period prior to test.

8. Beta reference source shall be in position as specified in note 3.

9. Test shall be performed with the assembly shown on figure 3 in a 1.4 mr/hr Cobalt-60 radiation field (calculated).

10. The tube shall be tested only with its longitudinal axis vertical and parallel to the direction of motion. The tube shall be mounted by means of a rigid clamp at the base of the tube.

11. The tube shall be tested only with its longitudinal axis horizontal and parallel to the direction of motion of the hammer. The tube shall be mounted by means of a rigid clamp at the base of the tube.

12. This test shall be performed at the conclusion of the holding period.

© 13. All tube windows shall be tested to assure that they can withstand an internal pressure of half of an atmosphere less than ambient pressure.

14. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

(C) 15. The shelf life requirements shall be as follows:

a. Tubes tested and procured in accordance with this tube specification sheet (TSS) shall be capable of a shelf life for a period of 18 months after the date of shipment from the manufacturer's plant. At anytime during this shelf life period the tubes shall be capable of meeting all requirements of quality conformance inspection, part 1, as specified herein. Tubes found to be defective within the shelf life period shall be replaced by the manufacturer at no cost to the Government.

b. A shelf life period of 90 days with the quality conformance inspection, part 1, starting voltage test shall be instituted during continuous production, but shipments of that month's production shall not be held pending completion of the test. Tubes shall be considered to conform to this TSS when three of the four tubes for each of the first 3 month's production and 75 percent of the cumulative quantity pass the test. If either of the above conditions are not satisfied, shipments shall be halted until three or four tubes of current production conform to the starting voltage test.

(C) 16. The relative plateau slope requirements shall be with the discrimination level of the scaler set at 0.25 V ±10 percent for standard Geiger-Mueller counter tube operation, and accomplished in accordance with the following steps:

a. With the equipment in operation and prepared for counting, set the high voltage adjust control to obtain 1,400 Vdc.

b. Mount standard reference source SBRS on the sample positioning shelf, place the shelf in position 1 of the tube, mount assembly and count for 5 minutes at 1,400 Vdc. A new Geiger-Mueller counter tube (one that has not been used for a month or more) should be counted for 15 minutes rather than 5 minutes. It is not necessary to record the count rate obtained in this step.

c. If the background is unknown, remove SBRS and obtain and record a 20 minute background count rate (CPM).

d. Place SBRS on the sample positioning shelf; place shelf in position 1 of the tube mount assembly.

CAUTION

If at any time while performing steps 16e through 16g it becomes apparent that the tube under test is in continuous discharge (rapidly increasing count rate for a constant value of applied voltage), immediately set high voltage adjust control to 1,200 Vdc. Determine if tube damage has occurred by raising the voltage in 50 Vdc increments every 5 minutes. Do not exceed a high voltage setting of 1,500 Vdc at any time. Otherwise continuous discharge will damage the tube. If continuous discharge occurs again, replace tube.

e. Count SBRS for 100,000 counts minimum in the same position at 1,300 Vdc and determine and record count rate in counts per minute (CPM). Count rate N1 equals total count divided by counting time in minutes minus background.

f. Count SBRS for 33,333 counts minimum in the same position at 1,400 Vdc and determine and record count rate in counts per minute. Count rate N2 equals total count divided by counting time in minutes minus background.

g. Count SBRS for 100,000 counts minimum in the same position at 1,500 Vdc and determine and record count rate in counts per minute. Count rate N3 equals total count divided by counting time in minutes minus background.

h. Determine plateau slope in percent as follows: Percent per 100 Vdc slope equals 100 times (N3 minus N1) divided by the quantity (N3 plus N1). This result shall be less than or equal to ±3 percent per 100 Vdc.

i. The measured count rate N2 shall be within ±10 percent of the quantity, (N1 plus N3) divided by 2.

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© 17. Requires Class A mandatory inspection, witnessed by DCAS/DCASR, on 100% basis.

© 18. Each tube shall be packaged in a container capable of withstanding pressures in the range of 87.3 to .760 mm of Hg.

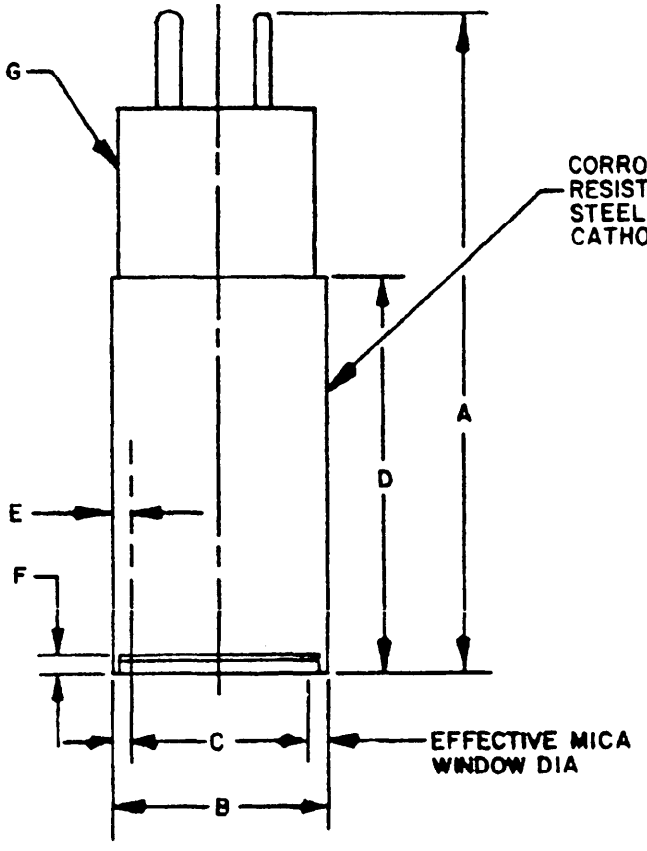
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Pin Connections	
Pin No.	Element
1	nc
2	a
3	nc
4	k

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.50 (114.3)
B	1.367 (34.72)	1.385 (35.18)
D	2.610 (66.29)	2.640 (67.06)
Quality conformance inspection, part 3 (See note)		
G	Base: A4-9 (EIA)	
Reference dimensions		
C	1.125 (28.58)	
E	0.109 (2.77)	
F	0.031 (.79)	

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 7840.

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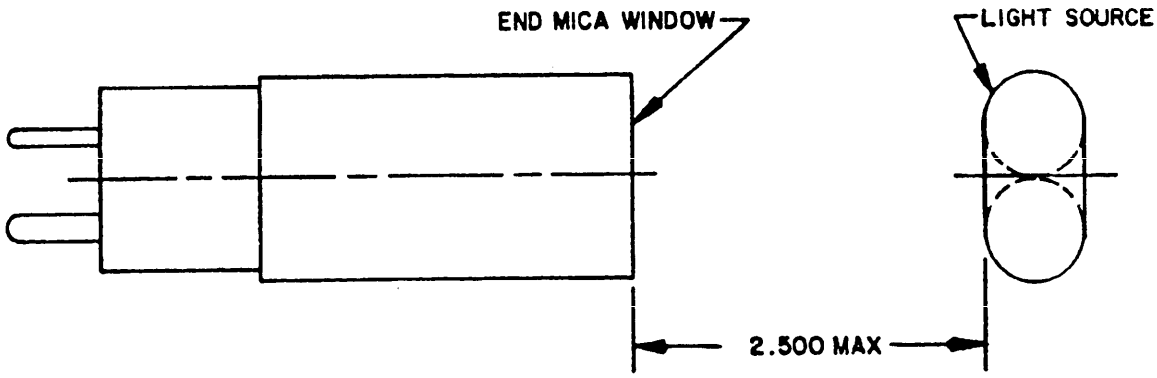


FIGURE 2. Position of tube for photosensitivity test.

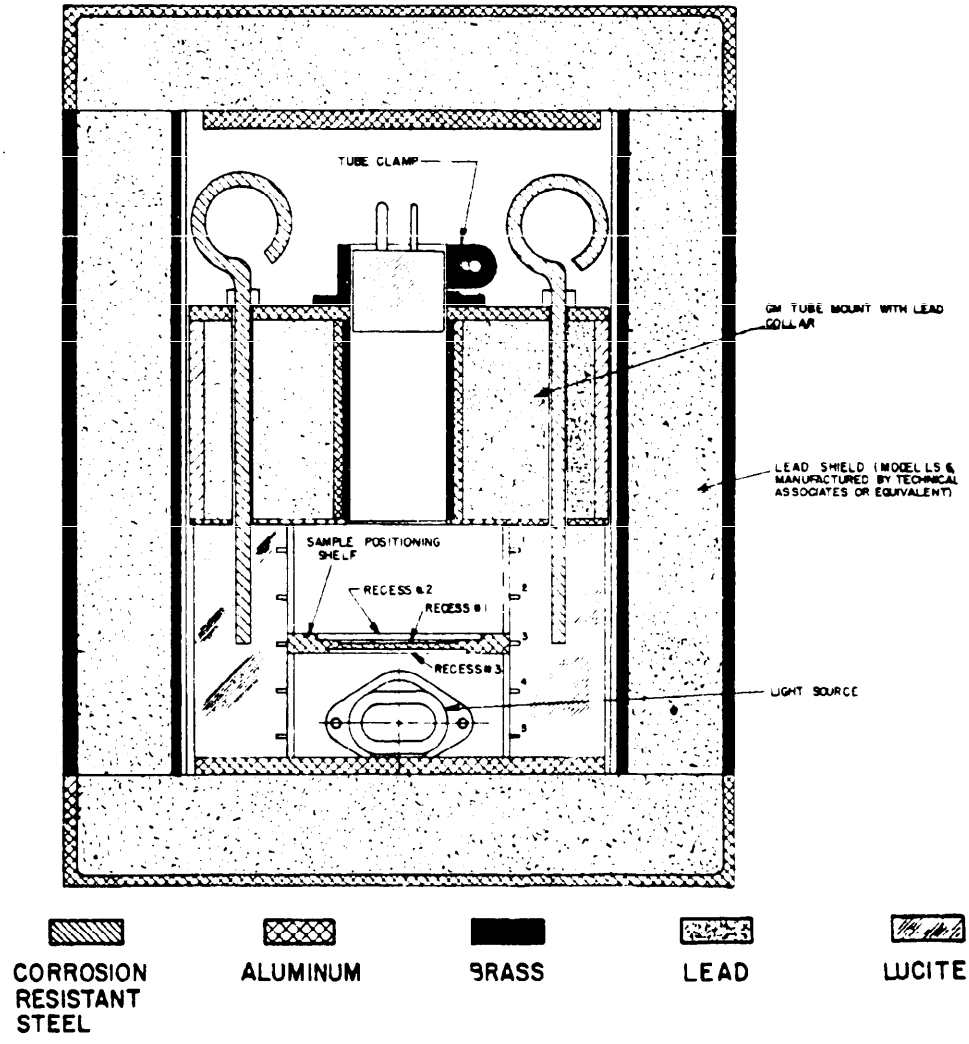


FIGURE 3. Tube position and shielding for response tests.

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